INCH-POUND

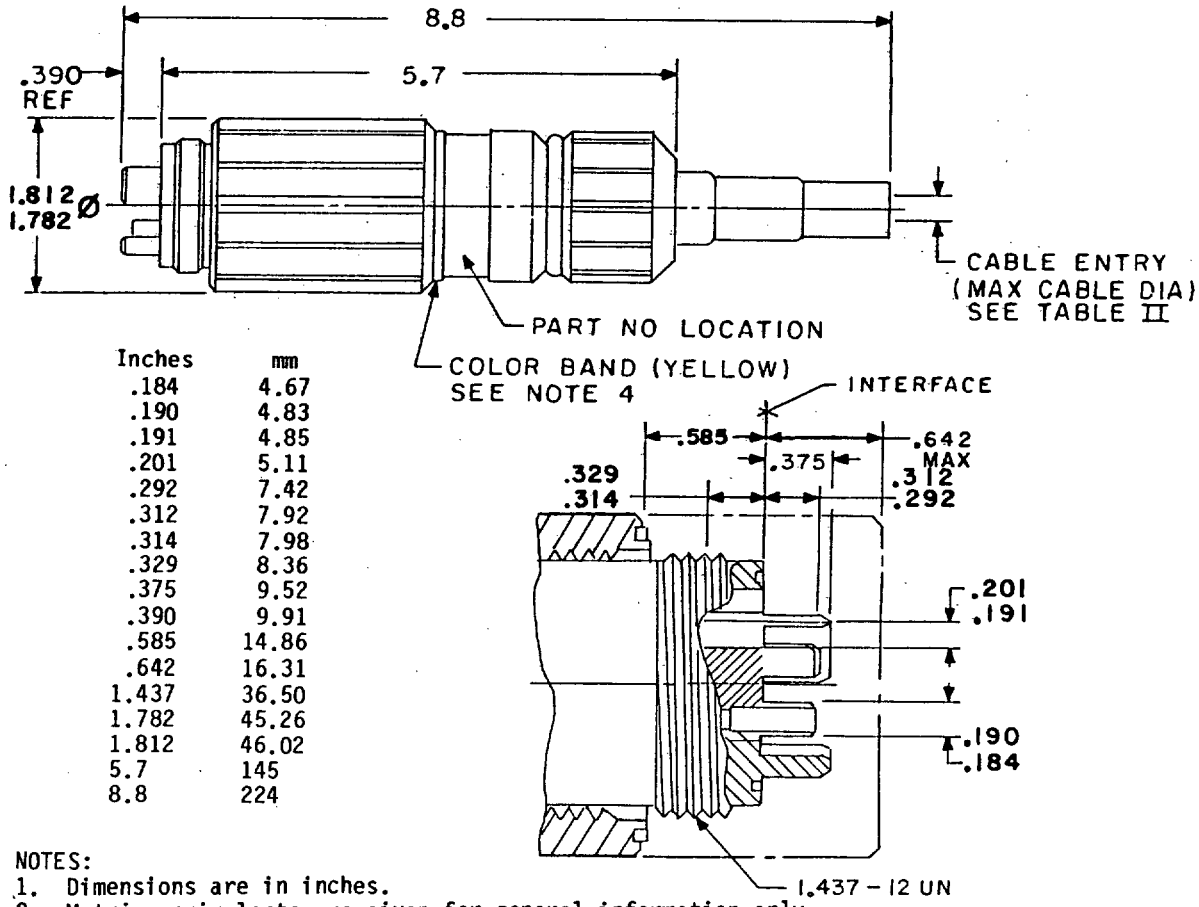
MIL-C-83526/2A
30 June 1989
SUPERSEDING
MIL-C-83526/2
19 August 1986

MILITARY SPECIFICATION SHEET

CONNECTORS, PLUG, FIBER OPTIC, CIRCULAR, HERMAPHRODITIC, IN-LINE MOUNTING, 6 POSITIONS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83526.



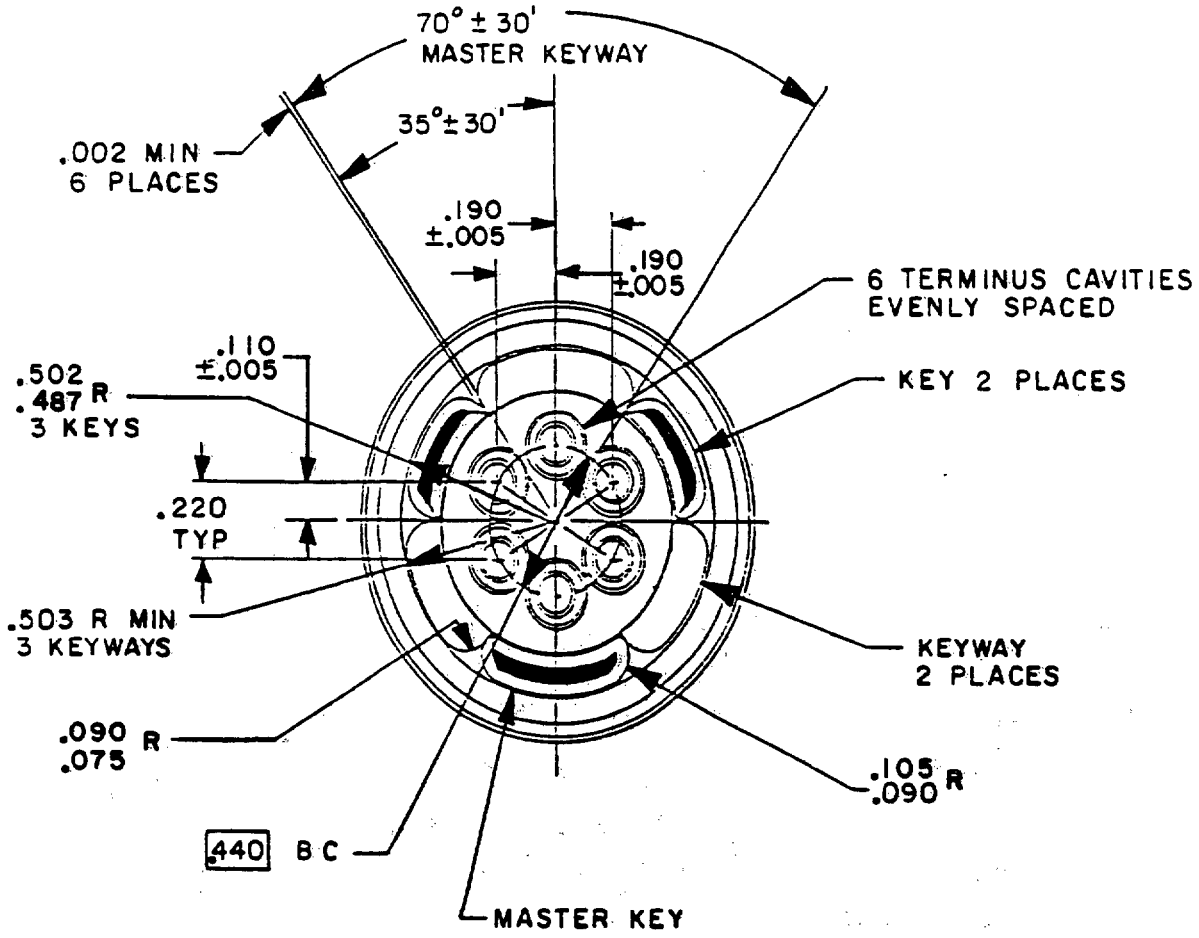
Inches	mm
.184	4.67
.190	4.83
.191	4.85
.201	5.11
.292	7.42
.312	7.92
.314	7.98
.329	8.36
.375	9.52
.390	9.91
.585	14.86
.642	16.31
1.437	36.50
1.782	45.26
1.812	46.02
5.7	145
8.8	224

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals, ±.03 (0.8 mm) for two place decimals and ±.1 (3 mm) for one place decimals. Angular tolerances are ±0° 30'.
4. The phrase "Fiber-Optic" must be included within the yellow band.
5. Unless otherwise specified, dimensions are symmetrical about centerlines.

FIGURE 1. In-line connector.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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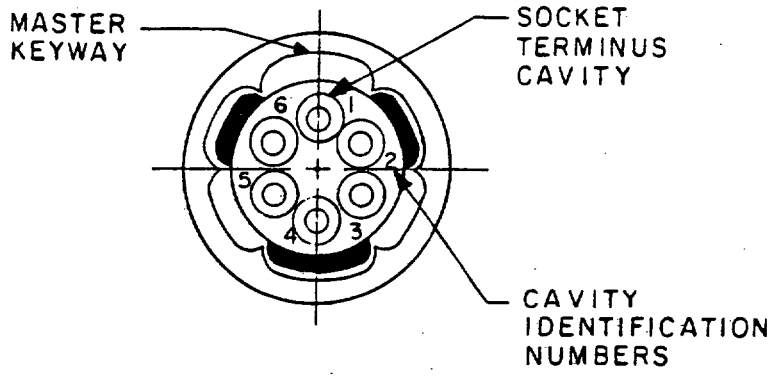


Inches	mm
.002	0.05
.005	0.13
.075	1.90
.090	2.29
.105	2.67
.440	11.18
.487	12.37
.502	12.75
.503	12.78

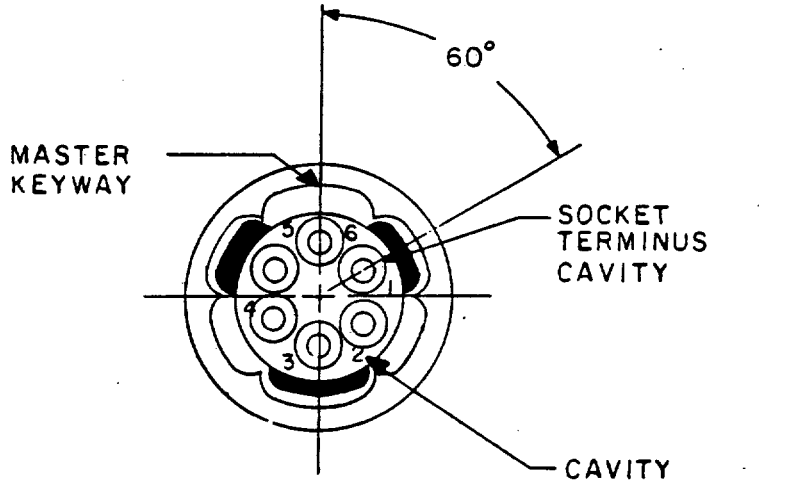
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Terminal configuration face view.

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KEYING POSITION A



KEYING POSITION B

NOTES: Cavities 1, 3, and 5 are socket terminus cavities.

FIGURE 3. Keying positions.

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1, 2, and 3.

Fiber optic cable:

Cable diameter: In accordance with DOD-C-85045.

Cable configuration: In accordance with DOD-C-85045.

Fiber diameter: In accordance with DOD-F-49291.

Fiber numerical aperture: In accordance with DOD-F-49291.

Fiber type: Multimode.

Fiber class: Graded index.

Materials: The material requirements are listed in table I.

Connector termination and assembly tools: As specified in MIL-K-83533.

Cleaning procedures: The cleaning procedures for the connector shall be included with the assembly instructions of the connector.

Termini: The required number of termini shall be supplied with each conector unless otherwise specified (see table II). Style P(pin) or S(socket), MIL-T-29504/1 and MIL-T-29504/2. For dummy terminus, see MIL-T-29504/3.

Weight: 15 ounces maximum.

Coupling forces: 11.0 pounds.

Coupling torques: 5 inch-pounds minimum.

Identification marking:

Part or Identifying Number (PIN): See table III.

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TABLE I. Material description.

Parts description	Material	Finish
Connector shell	Aluminum alloy 2011-T3	Anodized MIL-A-8625 type II, class 2 Color: Gray
Insert (6 termini)	QQ-A-225/3	
Compression nut	"	
Coupling nut and shroud	"	
Spacer	"	
Strain relief	"	Chemical film MIL-C-5541 class 1A
Compression sleeve	"	
Strain relief sleeve	"	
Jam nut	"	
Securing screw	Corrosion resistant Steel Type 303 SE, condition A. Per ASTM A581 or A582	Passivated MIL-S-5002
Dowel pin		"
Shaft retaining nut		"
Rotation stop ring		"
Rotation stop shaft		"
Retaining shaft	Beryllium copper QQ-C-530	None
Plug handle	Fluorosilicone rubber MIL-R-25988, class 1, grade 60, Color: Blue	None
Shrink tubing	Radiation crosslinked polyolefin MIL-I-23053/4, class 3	None
O-rings	Fluorosilicone rubber MIL-R-25988, class 1, grade 70, Color: Blue	None
Pistol seal	Molydisulfide-polyurethane MS28776	None

PIN:

M83526/02- 06 X X A, B, or N

Basic specification

Terminus insert arrangement

Key position (see figure 3)

Cable entry diameter (see table II)

Connector supplied with (A) 50/125 size termini, (B) 100/140 size termini, or (N) without termini

PIN example:

M83526/02 - 06A1B

The connector is a six channel connector supplied with the keying in position "A". The cable entry diameter is .256 inch maximum and the connector is supplied with the appropriate number of 50/125 termini.

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TABLE II. Cable entry diameter and PIN designation.

Cable entry diameter (over jacket)	Designation number
.256	1
.285	2
.307	3
.323	4

PIN: M83526/02- (dash number from table III). Marked on flange of connector.

TABLE III. PIN and characteristics.

PIN M83526/02-	Keying positions	Cable entry diameter	Fiber size DOD-F-49291	Cable PIN D85045	Termini supplied with connector
06A1A	A	.256	50/125	B6A	Yes
06A1B	A	.256	100/140	C6A	Yes
06A1N	A	.256	N/A	N/A	No
06B1A	B	.256	50/125	B6A	Yes
06B1B	B	.256	100/140	C6A	Yes
06B1N	B	.256	N/A	N/A	No
06A2A	A	.285	50/125	1/	Yes
06A2B	A	.285	100/140	1/	Yes
06A2N	A	.285	N/A	N7A	No
06B2A	B	.285	50/125	1/	Yes
06B2B	B	.285	100/140	1/	Yes
06B2N	B	.285	N/A	N7A	No
06A3A	A	.307	50/125	1/	Yes
06A3B	A	.307	100/140	1/	Yes
06A3N	A	.307	N/A	N7A	No
06B3A	B	.307	50/125	1/	Yes
06B3B	B	.307	100/140	1/	Yes
06B3N	B	.307	N/A	N7A	No
06A4A	A	.323	50/125	1/	Yes
06A4B	A	.323	100/140	1/	Yes
06A4N	A	.323	N/A	N7A	No
06B4A	B	.323	50/125	1/	Yes
06B4B	B	.323	100/140	1/	Yes
06B4N	B	.323	N/A	N7A	No

1/ D85045 cable not available for these connectors.

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Quality conformance and periodic inspection: The following cables shall be used for quality conformance and periodic inspection.

Number channels	Cables
6	PIN M85045/2-C6A, 100 micron core, 140 clad, or equal.

For quality conformance and periodic inspection, dummy termini shall be used in all unused cavities.

Patent notice: The Government does not have royalty-free license under the following patent for the benefit of manufacturers of the item, either for the Government or for use in equipment to be delivered to the Government.

Patent number	Patent expiration date
US 3,994,564	11/30/1993
US 4,054,366	10/18/1994
US 4,060,308	11/29/1994

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, SH
Air Force - 11, 17, 19, 80, 90, 99
DLA - ES

User activities:
Air Force - 13, 14

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0065-2)

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